



CaO–MgO–Al₂O₃–SiO₂ glass-ceramics from lithium porcelain clay tailings for new building materials



Weihong Zheng^{a,b,*}, Hua Cao^a, Jingbo Zhong^a, Shaoyang Qian^b, Zhigang Peng^b, Chunhua Shen^c

^a State Key Laboratory of Silicate Materials for Architectures, Wuhan University of Technology, 122 Luoshi Road, Wuhan, Hubei, PR China

^b Glass and Technology Research Institute of Shahe, Shahe, Hebei, PR China

^c The Center for Materials Research and Analysis, Wuhan University of Technology, Wuhan 430070, Hubei, PR China

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ABSTRACT

Lithium porcelain clay tailings (containing 1.2 wt.% Li₂O) blended with other chemical raw materials were used to develop a new glass-ceramic of CaO–MgO–Al₂O₃–SiO₂ system for rolling process. The effects of TiO₂ and ZnO oxides on melting and crystallization behaviors of glasses were investigated by DTA, XRD, EPMA and viscosity test. The results showed that melting temperature, forming temperature and glass transition temperature (*T_g*) of glasses were decreased with the increase of ZnO addition. Nevertheless, the crystallization temperature was shifted from 864 to 901 °C when the addition of TiO₂ decreased from 2 wt.% to 0.5 wt.%. Diopside as a major crystalline phase and spinel as a minor phase are precipitated in M3 specimen with 2.5 wt.% ZnO and 0.5 wt.% TiO₂. The columnar and granular crystalline phases were distributed in M3 specimen nucleated at 750 °C for 2 h and followed by crystallization at 1110 °C for 2 h, which resulted in proper appearance suitable for decorative materials.

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1. Introduction

There have been many researches on glass-ceramic materials in the quaternary system CaO–MgO–Al₂O₃–SiO₂ (CMAS) during previous years; it has focused on materials containing diopside as the main crystalline phase. CMAS glass-ceramics with diopside as a major crystalline phase generally have high strength together with high chemical resistance and have a wide variety of applications. According to researches, the glass-ceramics can be used as sealants for fuel cell applications [1], glazes for floor tiles [2], artificial bones, dental roots and crowns [3].

CMAS glass-ceramics are microcrystalline materials obtained from a parent glass by almost complete devitrification. The parent glasses could be produced by the traditional melting method from waste materials, such as thermal power plant fly ashes [4], fly ash from municipal solid wastes and bottom ash [5], vitrified waste [6]. Nevertheless, lithium porcelain clay tailings were rarely applied to the preparation of CMAS glass-ceramics.

In fact, Jiangxi Province in southeast China has rich lithium clay mineral resources. Therefore, a large number of tailings were produced in the process of mining, which resulted in serious environmental and health issues. The main components of lithium porcelain clay tailings are MgO, CaO, Al₂O₃, SiO₂, Li₂O, K₂O, Na₂O, Fe₂O₃, Cr₂O₃ and other oxides. According to the results of researches, lithium porcelain clay

tailings could be used as a raw material for CMAS glass-ceramics [4–6]. However, the introduction of Fe₂O₃ and Cr₂O₃ by lithium porcelain clay tailings led to dark color, which limits the application of glass-ceramics for building material [7]. Therefore, it is necessary to design a new CMAS decorating glass-ceramics with proper appearance for the utilization of lithium porcelain clay tailings.

In this work, the quaternary system CaO–MgO–Al₂O₃–SiO₂ glass-ceramics with lithium porcelain clay tailings are prepared by the traditional melting method. TiO₂ is used as a nucleant and ZnO as a flux. The effects of TiO₂ and ZnO on viscosity and crystallization of glass-ceramics are investigated by differential scanning calorimeter (DSC), X-ray diffraction (XRD) and electron probe microanalyzer (EPMA). The aim of present work is to produce a white glass-ceramic with low melting-temperature, which is suitable for rolling process and is a new decorative building material.

2. Experimental

2.1. Material preparation

Table 1 presents the chemical compositions of lithium porcelain clay tailings, which were evaluated by X-ray fluorescence (AxiosMAX, PANalytical) and chemical analysis. The (1.2 wt.% Li₂O) lithium porcelain clay tailings and the initial materials CaCO₃, MgCO₃, SiO₂, Al₂O₃, Na₂CO₃, K₂CO₃, TiO₂, ZnO and P₂O₅ were blended in various proportions to constitute the following chemical composition of parent glass (in wt.%), which are listed in Table 2. Glass batches (200 g) were mixed

* Corresponding author at: State Key Laboratory of Silicate Materials for Architectures, Wuhan University of Technology, 122 Luoshi Road, Wuhan, Hubei, PR China.

E-mail address: zhengweihong@whut.edu.cn (W. Zheng).

Table 1
Oxide composition (wt.%) of lithium porcelain clay tailings.

CaO	MgO	Al ₂ O ₃	SiO ₂	Na ₂ O	K ₂ O	Li ₂ O	P ₂ O ₅	Cr ₂ O ₃	Fe ₂ O ₃	Total
2.7	2.5	17.3	65.6	2.8	3.7	1.2	1.2	0.6	2.4	100.0

thoroughly, and then melted in a platinum crucible at 1350 °C for 4 h in an electric furnace. The melts were then poured into preheated steel molds and annealed at 600 °C for 30 min to relax internal stress. About 10 g of glass was quenched in water and dried in a dry box for the DSC measurement. The shaped glass was cut to a size of (30 × 10 × 5 mm) for heat treatment and test. Heat-treatment schedule of glass was determined by the results of the DSC curves. Glass samples were heat-treated under various conditions for the studies of phase transition and crystallization. After heat treatment, the samples were annealed at 700 °C for 30 min to relax internal stress.

2.2. Sample characterization

A mass of glass was selected and cut into pieces with the weight of about 10–50 g for each one. The samples were placed in correct quantity into the THETA Rheotronic II to measure the viscosity of glass above the softening point. The glasses were heated to a temperature that the viscosity of glass was lowered sufficiently to allow trapped air bubbles to be released. The viscosity (η ; in Pa·s) of glasses can be fitted to the Fulcher (FVT; Fulcher–Vogel–Tammann) equation:

$$\lg \eta = A + B/(T - T_0) \quad (1)$$

wherein, T is the temperature (°C) and A , B and T_0 are the adjustable constants.

Table 2
Oxide composition (wt.%) of CMAS glass-ceramics.

Sample	CaO	MgO	Al ₂ O ₃	SiO ₂	Na ₂ O	K ₂ O	Li ₂ O	ZnO	P ₂ O ₅	TiO ₂	Fe ₂ O ₃	Cr ₂ O ₃	Total
M1	17.4	8	22	36	8	2	0.6	1	1.5	2	1.2	0.3	100
M2	17.4	8	22	36	8	2	0.6	2	1.5	1	1.2	0.3	100
M3	17.4	8	22	36	8	2	0.6	2.5	1.5	0.5	1.2	0.3	100

Crystallization kinetics of the glasses was studied using differential scanning calorimeter (DSC). DSC of quenched glass was carried out in a Netzsch DSC 404F3. After crushing quenched glass to about 100–200 μ m, non-isothermal experiments were performed by heating 30 mg samples in a Pt crucible with Al₂O₃ as the reference material in the temperature range between 20 and 1200 °C with a heating rate of 5–20 °C/min. The variation of the crystallization peak positions with different DSC heating rates can be used to estimate the activation energy for crystallization and to determine the crystallization mechanism. The crystallization behavior of glasses was determined by the Kissinger and Augis–Bennet equations [8,9].

The heat treated samples were subjected to powder X-ray diffraction (XRD) using a Rigaku Ultima IV X-ray Diffractometer (Tokyo, Japan) with Cu K α radiation in the 2 θ range from 10 to 70° at steps of 0.02°.

The fine microstructure of the glasses was observed by an electron probe microanalyzer (EPMA) with a Jeol JXA-8230. The heat-treated samples were etched with 5% HF solution at room temperature for 1 min before the EPMA analysis. The samples were washed ultrasonically with ethanol for 5 min after etching. Etched glass samples were coated with a thin layer of carbon.

3. Results

3.1. The viscosity of glasses

Glass viscosity is one of the key properties for melting, fining, processing optimization, glass formation and annealing process. Rolling process requires a very precise control of the viscosity throughout the glass melting and forming process in order to achieve high throughput and high yield of acceptable products.

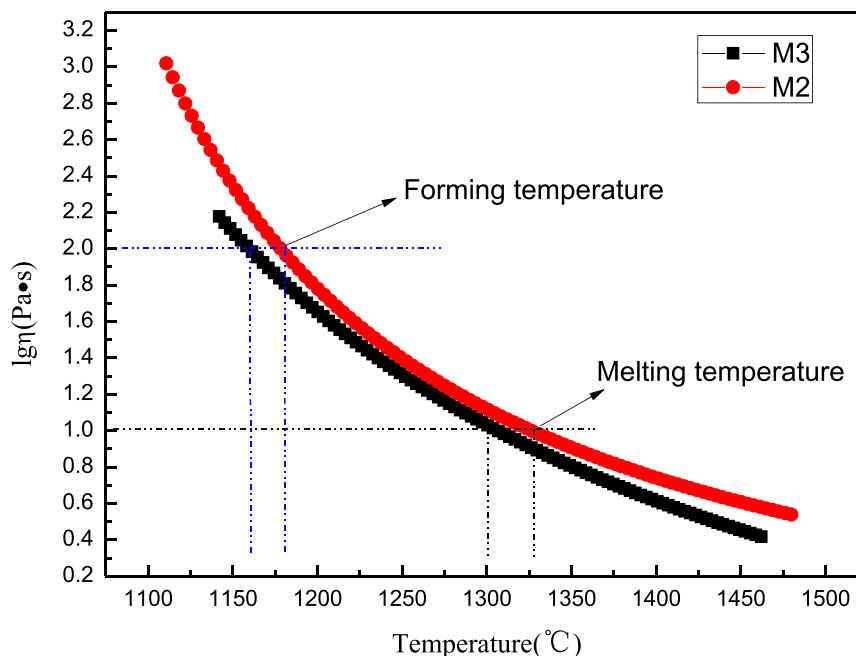


Fig. 1. Viscosity–temperature curves of M2 and M3.

Table 3

Parameters for VFT equation and four characteristic temperature intervals of forming process for glass samples.

Sample code	M2	M3
<i>VFT equation parameter</i>		
A	−0.62885	−1.50605
B	634.37057	1289.25702
T_0	936.93142	792.05309
R^2	0.99117	0.99876
<i>Characteristic temperature (°C)</i>		
Melting point ($\lg \eta = 1$)	1326.4	1306.5
Forming point ($\lg \eta = 2$)	1178.2	1159.8
Working point ($\lg \eta = 3$)	1111.7	1078.2
End of forming process ($\lg \eta = 5.75$)	1036.4	969.7

The viscosity of the glass specimens is plotted as a function of temperature in Fig. 1. It is the natural behavior of glass for the decrease of viscosity as temperature increases; however, the decreasing rate can be changed by the content of TiO_2 and ZnO . The curves in Fig. 1 show that the viscosity of M3 is lower than M2 during the temperature region of 1100–1480 °C. Furthermore, temperature-ranges for typical glass-making operations are decreased by the increase of ZnO addition, which are shown in Table 3. As a result, the melting temperature, forming temperature and working temperature of M3 are decreased to 1306.5, 1159.8 and 1078.2 °C, respectively.

3.2. Influence on crystallization

Fig. 2 shows the DSC curves of the CMAS glasses. The DSC curve of M1 specimen shows a small endothermic peak valley at 725 °C, a major exothermic peak at 864 °C and a shoulder peak at 943 °C. And the DSC trace of M2 specimen exhibits an endothermic peak valley at 720 °C, a major peak at 873 °C and a shoulder peak at 951 °C. The first peak implies the glass transition temperature (T_g), and the second and third peaks could be attributed to the crystallization (T_{p1}) or transformation of crystalline phase (T_{p2}) as discussed in literatures.

However, the DSC trace of M3 specimen has only an endothermic peak valley at 717 °C and a board exothermic peak at 901 °C, without other peaks, which indicates that T_{p1} and T_{p2} peaks have joined. Therefore, the temperature of crystallization was shifted to an intermediate temperature and crystallization reactions occurred at closer temperatures. The T_g of CMAS glasses shifts to lower temperature as the addition of ZnO is increased and the TiO_2 level is decreased, which shows that ZnO decreases the crystal nucleation temperature. Moreover, the

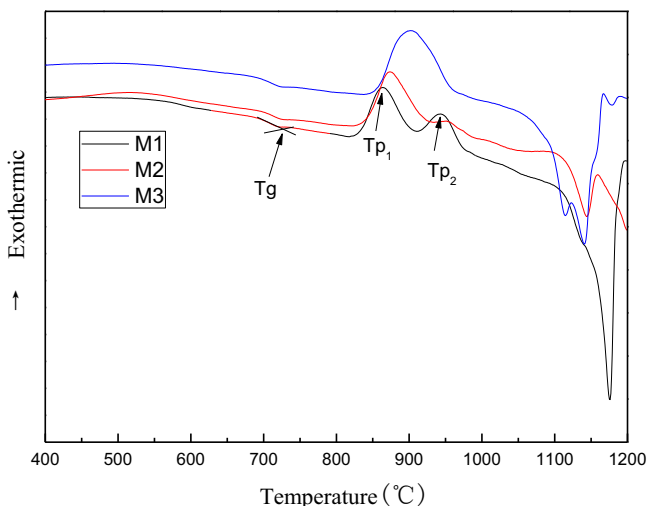


Fig. 2. The DSC curve of CMAS initial glass at 10 °C/min.

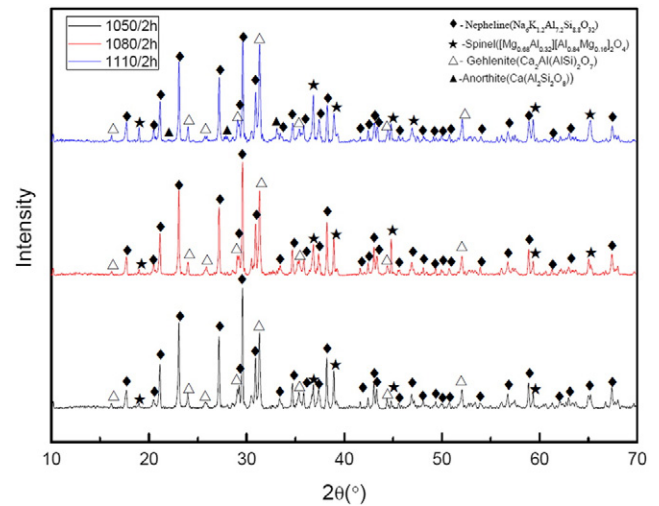


Fig. 3. The XRD results of M1 specimen treated at 750 °C for 2 h and at 1050, 1080 and 1110 °C for 2 h.

crystallization peak of M3 is shifted to higher temperature when the addition of TiO_2 is decreased simultaneously with ZnO level increased. The crystallization peaks (T_p) on the DSC curves imply that crystalline phase forms during the heat-treatment, which are also confirmed by XRD results.

Figs. 3, 4 and 5 show the diffraction patterns of samples treated at different temperatures for 2 h. In Fig. 3, the XRD pattern of M1 specimen treated at 1050 °C for 2 h shows nepheline ($\text{Na}_6\text{K}_{1.2}\text{Al}_{7.2}\text{Si}_{8.8}\text{O}_{32}$, JCPDS-PDF 85-1487, $a = b = 10.007$ Å, $c = 8.385$ Å), spinel ($(\text{Mg}_{0.68}\text{Al}_{0.32})(\text{Al}_{0.84}\text{Mg}_{0.16})_2\text{O}_4$, JCPDS-PDF 96-0083, $a = b = c = 8.084$ Å) and gehlenite ($\text{Ca}_2\text{Al}(\text{AlSi})_2\text{O}_7$, JCPDS-PDF 89-5917, $a = b = 7.717$ Å, $c = 5.086$ Å) were precipitated. As the temperature increased from 1050 to 1110 °C, the crystalline phases were still nepheline, gehlenite and spinel. However, the new phase anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$, JCPDS-PDF 76-0948, $a = 8.184$ Å, $b = 12.890$ Å, $c = 14.214$ Å) was precipitated in M1 specimen treated at 1110 °C. Furthermore, the content of anorthite phase was lower than nepheline and spinel phases.

The XRD patterns of M2 specimen treated at 1050, 1080 and 1110 °C for 2 h (Fig. 4) show that nepheline and akermanite are precipitated in these specimens. As the temperature increases from 1050 to 1110 °C,

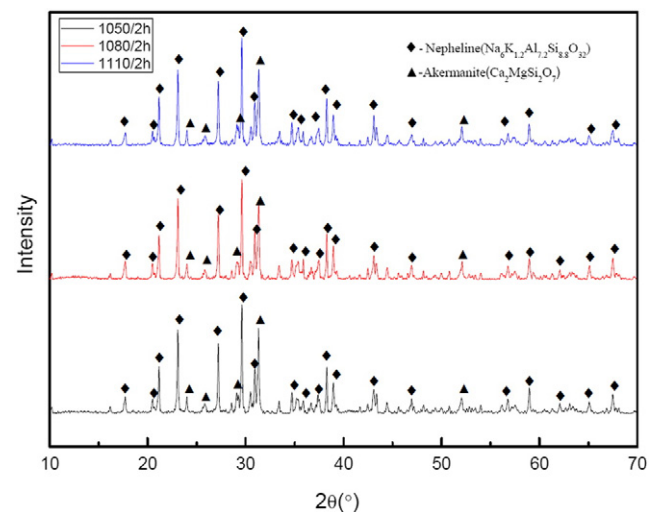


Fig. 4. The XRD results of M2 specimen treated at 750 °C for 2 h and at 1050, 1080 and 1110 °C for 2 h.

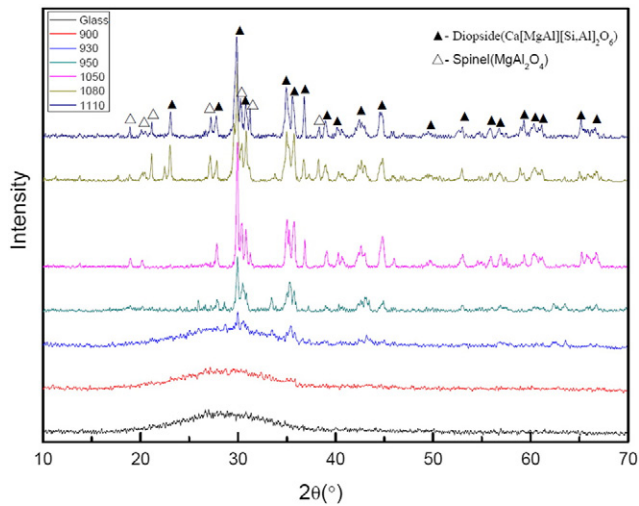


Fig. 5. The XRD results of M3 specimen treated at 750 °C for 2 h and at 900, 930, 950, 1050, 1080 and 1110 °C for 2 h.

the content of akermanite phase is gradually increased according to the intensity of peaks.

According to the XRD patterns of Fig. 5, M3 parent glass has a broad scattering peak up to 930 °C, which shows that the specimen is an amorphous solid. As temperature was higher than 930 °C, the broad scattering spectrum disappeared and diopside ($\text{Ca}(\text{MgAl})(\text{SiAl})_2\text{O}_6$, JCPDS-PDF 41-1370, $a = 9.732 \text{ \AA}$, $b = 8.867 \text{ \AA}$, $c = 5.279 \text{ \AA}$) as a major crystalline phase is precipitated. The content of diopside phase increases gradually when temperature is increased from 930 to 1110 °C. However, spinel (MgAl_2O_4 , JCPDS-PDF 99-0098, $a = b = c = 8.083 \text{ \AA}$) is precipitated in the specimen as the minor crystalline phase when temperature is higher than 950 °C.

3.3. Microstructure analysis

Fig. 6 shows the SEM/EPMA results of CMAS glass-ceramics treated at 1110 °C for 2 h. According to the SEM image of M1, a large number of granular crystals and a small quantity of lamellar shape crystal were distributed in the glass matrix after treated at 1110 °C. Therefore, it can be found that nepheline and spinel with granular shape and gehlenite with lamellar shape are precipitated according to the XRD results of M1 specimen. According to the SEM image and XRD patterns of M2 specimen, nepheline and akermanite with lamellar shape were precipitated.

In Fig. 7, the columnar and small granular crystalline phases were distributed in M3 specimen after crystallization at 1110 °C. It is well known that spinel is a granular crystal with an octahedron shape and

diopside is a columnar crystal in glass-ceramics [10,11]. Furthermore, the EDS microanalyses were carried out on different areas of M3 specimen. The magnesium content of A area is higher than other areas. Combined with the results of XRD patterns, it could be found that these granular crystalline phases would be spinel. The aluminium and oxygen content of B and C areas is apparently higher than A area, which showed that the columnar crystalline phases were diopside.

According to the results of EPMA, it can be found that the grain size of M3 sample is higher than M1 and M2 samples; however, the major crystalline phase of specimens is altered by TiO_2 and ZnO . The EPMA results demonstrated that the increase of ZnO and the decrease of TiO_2 promoted the formation of diopside phase simultaneously with precipitation of a small quantity of spinel.

4. Discussions

4.1. The effects of ZnO and TiO_2 on the viscosity of glasses

The importance of viscosity as a property of molten glasses has long been recognized. For practical purposes, a number of important stages in the variation of viscosity with temperature have been named; these names serve to label the temperatures at which particular glass-making operations can be carried out. The temperatures of M2 and M3 specimens at different viscosities are shown in Table 3. The temperature for the viscosity of $10 \text{ Pa}\cdot\text{s}$ is referred to as the melting temperature of glass, so the melting temperature of M2 and M3 is 1326 and 1306 °C, respectively. It indicates that the addition of ZnO and TiO_2 can alter melting temperature and viscosity. The valence, co-ordination, ionic radius, single bond strength of Si^{4+} , P^{5+} , Zn^{2+} and Ti^{4+} are shown in Table 4. Table 4 shows that P^{5+} and Si^{4+} have high bond energy and covalency, which will strengthen the polymerization of glass network [12]. However, Ti^{4+} as intermediates has high co-ordination and low bond energy, which make it possible to enter into the glass network and results in the compact connection of glass network. Therefore, the addition of TiO_2 increases the viscosity of glass melt. Nevertheless, the bond energy of Zn^{2+} is only 36 kcal and similar to Na^+ , which shows that ZnO is the modifier in glass network and Zn^{2+} ions are located at network interspace. As a result, the polymerization of glass network is extremely destroyed and viscosity is decreased with ZnO addition.

4.2. The effects of ZnO and TiO_2 on crystallization kinetics of glasses

The crystallization kinetic characteristics of CMAS system glass can be determined as the following by Kissinger [8] and Augis–Bennet [9], which are, respectively, expressed as

$$k = v \exp\left(-\frac{E}{RT}\right) \quad (2)$$

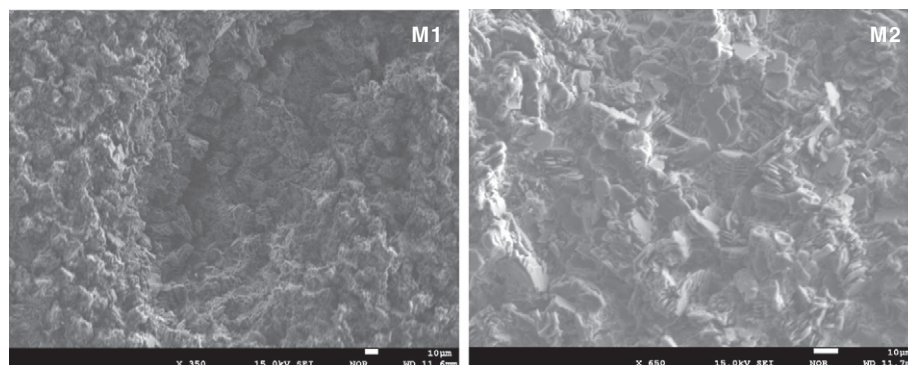


Fig. 6. SEM micrographs of the M1 and M2 samples nucleated at 750 °C for 2 h followed by crystallization at 1110 °C for 2 h.

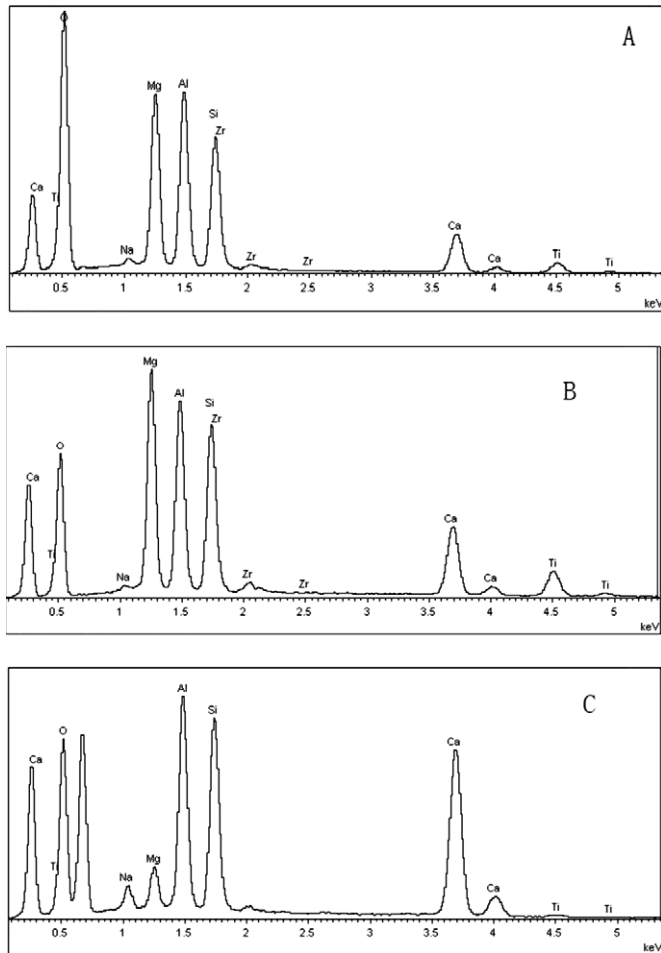
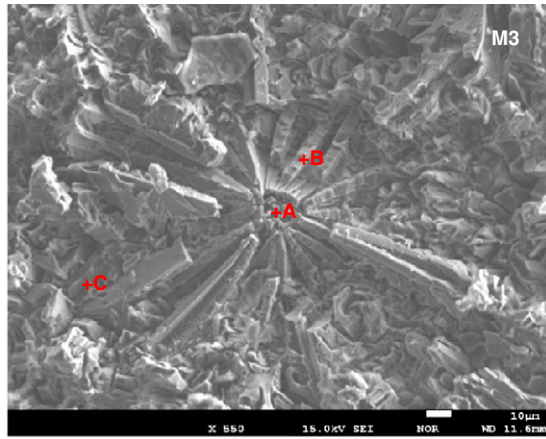


Fig. 7. SEM micrographs and EDS of the M3 specimen nucleated at 750 °C for 2 h followed by crystallization at 1110 °C for 2 h.

$$\ln \left(\frac{T_p^2}{\alpha} \right) = \frac{E}{RT_p} + \ln \frac{E}{R} - \ln v \quad (3)$$

$$n = \frac{2.5}{\Delta T} \times \frac{RT_p^2}{E} \quad (4)$$

wherein, E is the effective activation energy of crystal growth (kJ/mol), R is the universal gas constant, v is the effective frequency factor, α is the DSC heating rate (°C/min), k is the reaction rate constant, which is

Table 4

Various parameters of Si^{4+} , P^{5+} , Zn^{2+} , Ti^{4+} and Zr^{4+} .

M in MO_x	Valence	Co-ordination	Ionic radius (pm)	Single bond strength $B_{\text{M-O}}$ (kcal)
<i>Glass formers</i>				
Si	4	4	42	106
P	5	4	34	88–111
<i>Intermediates</i>				
Ti	4	6	74.5	73
<i>Modifiers</i>				
Zn	2	4	74	36
Na	1	6	116	20

related to E and v , n is the crystallization kinetic parameters, i.e. Avrami parameter, and ΔT is the half-height temperature width of the maximum exothermic peak of DSC curve. The crystallization kinetic parameter n is determined by the nucleation process and growth morphology. The crystallization mechanism is surface crystallization when n value is close to 1, two-dimension crystallization when n is close to 2 and volumetric crystallization when n is close to 3 [13].

The DSC curves of M1, M2 and M3 specimens treated at 5, 10, 15 and 20 °C/min are shown in Fig. 8. Table 5 shows the crystallizing peak temperatures (T_p) from DSC curves. According to Eqs. (2), (3) and (4), the relationship between $\ln(T_p^2/\alpha)$ and $1/T_p$ is constructed (Fig. 9) to calculate the effective activation energy (E), frequency factor (v), crystal index (n) and reaction rate constant (k). The results are shown in Table 6.

According to the XRD results, nepheline and akermanite were precipitated in M1 and M2 specimens treated at the temperature above T_p . It can be observed that the increase of 1 wt.% ZnO addition instead of TiO_2 decreases E from 386.61 to 382.27 kJ/mol. Furthermore, the n value was decreased from 3.86 to 3.85 as the increase of ZnO addition. However, the v and k values were shifted to $2.90 \times 10^{17} \text{ min}^{-1}$ and 1.092, respectively. The results show that the crystallization was promoted with the increase of ZnO addition. The effect of ZnO on microstructure and crystallization of glass with composition $25\text{SiO}_2\text{--}50\text{CaO--}15\text{P}_2\text{O}_5\text{--}(10-x)\text{Fe}_2\text{O}_3\text{--}x\text{ZnO}$ has been studied by K. Sharma [14]. Results have shown a decrease in the fraction of non-bridging oxygen with an increase in zinc oxide content. The formation of thin flake-like structure with coarse particles was observed at high zinc oxide concentration [14].

The diopside as a major crystalline phase and spinel as a minor phase were precipitated in M3 specimen treated at 950–1100 °C according to the XRD curves in Fig. 5. It could be found that the effective activation energy (E) of M3 was only 255.75 kJ/mol and the n value was 3.57, which were lower than those of M1 and M2 specimens. The reason for different activation energies of M1, M2 and M3 was that the energy for diopside precipitation was lower than nepheline. According to the research by M.L. Öveçoğlu, diopside as a major crystalline phase was precipitated in the $\text{CaO--MgO--Al}_2\text{O}_3\text{--SiO}_2$ system from waste and the activation energy for crystallization value was 150 kJ/mol [15]. Furthermore, the increase of ZnO addition and the decrease in TiO_2 -level change the crystallization mechanism from volumetric crystallization to two-dimension crystallization, which is in accordance with the conclusions drawn by Matusita and Sakka [16].

As a result, the columnar and granular crystalline phases were distributed in M3 specimen nucleated at 750 °C for 2 h followed by crystallization at 1110 °C for 2 h. Fig. 10 showed that M3 specimen treated at 1100 °C has an interesting appearance suitable for decorative materials.

5. Conclusions

In the present work, the lithium porcelain clay tailings blended with other raw chemical materials were used to develop a new glass-ceramic of $\text{CaO--MgO--Al}_2\text{O}_3\text{--SiO}_2$ system suitable for rolling process and

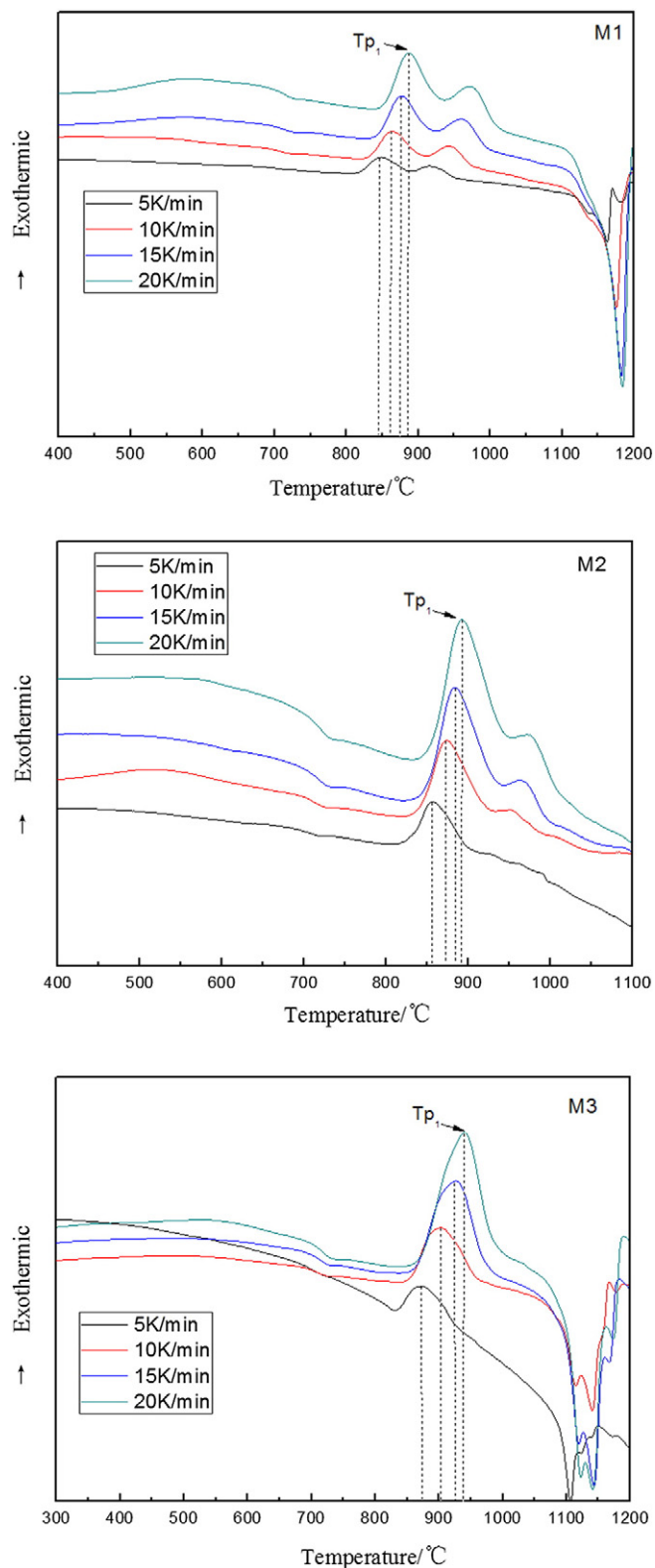


Fig. 8. The DSC curve of CMAS initial glass at 5, 10, 15 and 20 °C/min.

Table 5

T_g , T_{p1} and T_{p2} temperatures (°C) of all specimens at 10 °C/min.

	T_g	T_{p1}	T_{p2}
M1	725	864	943
M2	720	873	951
M3	717	901	–

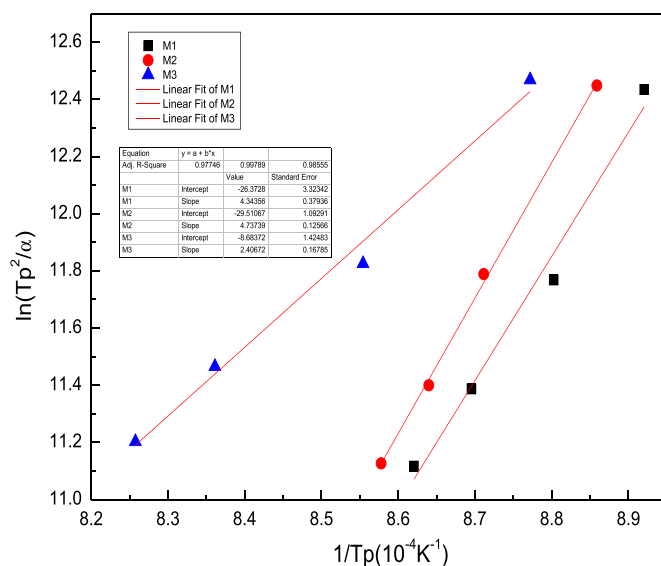


Fig. 9. Relationship between $\ln(T_p^2/\alpha)$ and $1/T_p$.

decorative building materials. Based on the results reported in the investigation, the conclusions can be drawn as follows:

- (1) The melting temperature, forming temperature and glass transition temperature (T_g) of M3 glass with 2.5 wt.% ZnO and 0.5 wt.% TiO₂ were 1306, 1160 and 717 °C, respectively. The low melting temperature resulted in the low energy-consumption and the proper forming temperature was suitable for rolling process.
- (2) The crystallization temperature was shifted from 864 to 901 °C when the addition of TiO₂ decreased from 2 wt.% to 0.5 wt.%. Nepheline and akermanite were precipitated in M1 and M2 specimens treated at 1050 °C. As the temperature increases from 1050 to 1110 °C, the crystalline phases of M2 specimen were still nepheline and akermanite and the crystallinity increased gradually. However, the crystalline phase changed when the heat treatment temperature was further increased.
- (3) For heat treatment temperature lower than 930 °C, the M3 specimen was glassy. Diopside as a major crystalline phase and spinel as a very minor phase were precipitated in M3 specimen treated at 950 °C. The content of crystalline phase increased gradually when heat treatment temperature increased from 950 to 1110 °C.
- (4) The columnar and granular crystalline phases were distributed in M3 specimen nucleated at 750 °C for 2 h followed by crystallization at 1110 °C for 2 h, such as the glass ceramic resulted in proper appearance suitable for decorative materials.

Table 6

E , n , v and k crystallization values of M1, M2 and M3 specimens.

Sample no.	M1	M2	M3
$\alpha = 5$ °C/min	$T_p = 850$ °C	$T_p = 855$ °C	$T_p = 880$ °C
$\alpha = 10$ °C/min	$T_p = 864$ °C	$T_p = 873$ °C	$T_p = 901$ °C
$\alpha = 15$ °C/min	$T_p = 876$ °C	$T_p = 884$ °C	$T_p = 921$ °C
$\alpha = 20$ °C/min	$T_p = 887$ °C	$T_p = 893$ °C	$T_p = 938$ °C
E (kJ/mol)	386.61	382.27	255.75
n ($\alpha = 10$ °C/min)	3.86	3.85	3.57
v (min ⁻¹)	3.91×10^{15}	2.90×10^{17}	4.82×10^{10}
k	0.007	1.092	0.201



Fig. 10. Visual appearance of M3 specimen nucleated at 750 °C for 2 h followed by crystallization at 1110 °C for 2 h.

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